94.3
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Shubham Singh BHadoria

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94.3% • 165.0 / 175
scored in eClerx - Java Coding Test - New in 41 min 8 sec on 12 Mar 2026 19:26:25 IST

Candidate Information

Email	sbhadoria787@gmail.com
Test	eClerx - Java Coding Test - New
Candidate Packet	View
Taken on	12 Mar 2026 19:26:25 IST
Time taken	41 min 8 sec/ 90 min
Work Experience	5 years
Invited by	Vipul
Invited on	11 Mar 2026 18:53:15 IST

Skill Distribution

No.	Skill	Score
1	Spring Boot Advanced	50%

2	Spring Boot Intermediate	66%
3	Problem Solving Intermediate	100%
4	Java Intermediate	100%

Tags Distribution

Spring Boot	60%	Hard	50%
TDD	100%	Medium	96%
Exception Handling	0%	Aspect-Oriented Programming	0%
Graphs	100%	Greedy	100%
Sorting	100%	Java	100%

Questions

Spring Boot (Advanced) • 15 / 25

No.	Question	Skill	Score	Code Quality	Time Taken	Status
1	JSON Deserialization Multiple Choice	Spring Boot (Advanced)	0/5	-	1 min 32 sec	⊗

2	Response Multiple Choice	Spring Boot (Advanced)	5/5	-	2 min 3 sec	✓
3	Endpoint Security Multiple Choice	Spring Boot (Intermediate)	5/5	-	1 min 6 sec	✓
4	CSRF Multiple Choice	Spring Boot (Intermediate)	5/5	-	1 min 2 sec	✓
5	Predict the Output Multiple Choice	Spring Boot (Intermediate)	0/5	-	2 min 39 sec	✗

Problem Solving (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status
6	Minimize Path Value Coding	Problem Solving (Intermediate)	75/75	-	21 min 39 sec	✓

Java (Intermediate) • 75 / 75

No.	Question	Skill	Score	Code Quality	Time Taken	Status
7	Java: DataVille's Optimal Storage Coding	Java (Intermediate)	75/75	-	9 min 36 sec	✓

1. JSON Deserialization

✗ Incorrect

Multiple Choice | Spring Boot | Hard

Question description

CheckerCreateRequest.java

```
// Annotation here
public class CheckerCreateRequest {
    private String name;
    private String surname;
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getSurname() {
        return surname;
    }
    public void setSurname(String surname) {
        this.surname = surname;
    }
}
```

CheckerController.java

```
@RestController
@RequestMapping("/checkers")
public class CheckerController {

    @Autowired
    CheckerService checkerService;

    @PostMapping
    public ResponseEntity createChecker(@RequestBody CheckerCreateRequest checkerCreateReque
st) {
        checkerService.createChecker(checkerCreateRequest);
        return new ResponseEntity(HttpStatus.CREATED);
    }
}
```

Request Json data

```
{
   "checker":{
      "name": "Checker",
      "surname":"Number2"
   }
}
```

A POST request with "Request Json data" is made to "/checkers". What annotation needs to be added to the "CheckerCreateRequest" class so that it can successfully deserialize and process the request?

Interviewer guidelines

@JsonRootName doesn't add checker as a wrapper attribute by itself. To make it successfully deserialize with given json annotations on D should be used together.

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ● **@JsonRootName("checker")**
- ○ @JsonName("checker")
- ○ @JsonTypeName("checker")
- ○ @JsonTypeInfo(include = JsonTypeInfo.As.WRAPPER_OBJECT, use = JsonTypeInfo.Id.NAME) @JsonTypeName("checker") ✓

No comments.

2. Response

Correct

Multiple Choice | Hard | Spring Boot | TDD

Question description

Which response will cause the following test case to fail?

```
@Test
   @DisplayName("Filter Price Check 3")
   public void FilterPriceCheck3() throws Exception {
      mockMvc.perform(get("/filter/price/150/500"))
            .andDo(print())
            .andExpect(jsonPath("$[*].product", containsInAnyOrder("Product 1", "Product 2" , "Produc
t 5")))
   }
```

Candidate's Solution

Options: (Expected answer indicated with a tick)

- [ ] [{"product" : "Product 5"} , {"product" : "Product 1"}, {"product" : "Product 2"}]
- [ ] [{"product" : "Product 5"} , {"product" : "Product 2"}, {"product" : "Product 1"}]
- [ ] [{"product" : "Product 1"} , {"product" : "Product 2"}, {"product" : "Product 5"}]

[{"product" : "product 5"} , {"product" : "product 2"}, {"product" : "product 1"}]

No comments.

3. Endpoint Security

Correct

Multiple Choice | Spring Boot | Medium

Question description

In Spring Boot Security, the option to enable secure endpoints with @PreAuthorize annotation is:

Interviewer guidelines

@EnableGlobalMethodSecurity by default doesn't enable this feature. To enable this prePostEnabled=true value must be set

Candidate's Solution

Options: (Expected answer indicated with a tick)

- Configuration class annotated with @EnableGlobalMethodSecurity
- Configuration class annotated with @EnableGlobalMethodSecurity(securedEnabled = true)
- **Configuration class annotated with @EnableGlobalMethodSecurity(prePostEnabled = true)**

None above

No comments.

4. CSRF

Correct

Multiple Choice | Spring Boot | Medium

Question description

Which Java Spring-based code segment correctly configures CSRF?

Interviewer guidelines

CSRF Protection makes sure that server is not bombarded with unwanted calls. Spring Security enables some features by default like CSRF and Form based authentication.

Candidate's Solution

Options: (Expected answer indicated with a tick)

private void manage(HttpSecurity http) {   http.csrf().allowed() }

private void manage(HttpSecurity http) {   http.csrf() }

private void manage(HttpSecurity http) {   http.csrf().enable() }

None of the above. CSRF is configured by default when Spring Security Dependency is added.

No comments.

5. Predict the Output

Incorrect

Multiple Choice | Medium | Exception Handling | Spring Boot | Aspect-Oriented Programming

Question description

Consider the following Snippet and predict the output.

The following API is triggered: http://localhost:8080/add/employee?eId=123&fName=Ravi&lastName=Shekhar

```
@Autowired
private EService eService;

@RequestMapping(value = "/add/employee", method = RequestMethod.GET)
public com.model.Employee add(@RequestParam("eId") String eId, @RequestParam("fName") Strin
g fName, @RequestParam("lastName") String lastName)
{
   return eService.createEmployee(eId, fName, lastName);
}

@Aspect
@Component
public class EServiceAspect
{
   @After(value = "execution(* com.random.service.EService.*(..)) && args(eId, fName, lastName)")
```

```
    public void afterAdvice(JoinPoint joinPoint, String eId, String fName, String lastName) {
        System.out.println("After method:" + joinPoint.getSignature());
        System.out.println("Creating Employee with first name - " + fName+ ", second name - " + lastNa
me+ " and id - " + eId);
    }
}
```

Candidate's Solution

Options: (Expected answer indicated with a tick)

- ○ After method : com.test.model.Employee addEmployee(String,String,String) Creating Employee with first name - Ravi , second name - Shekhar and id - 123

- ● ** After method : com.test.model.Employee addEmployee(string,string,string) Creating Employee with first name - Ravi , second name - Shekhar and id - 123   **

- ○ After method : com.test.model.Employee addEmployee(empId,firstName,secondName) Creating Employee with first name - Ravi , second name - Shekhar and id - 123

- ○ After method : com.test.model.Employee addEmployee(String,String,String) Creating Employee with first name - Shekhar, second name - Ravi and id - 123

No comments.

6. Minimize Path Value

Correct

Coding Graphs Medium

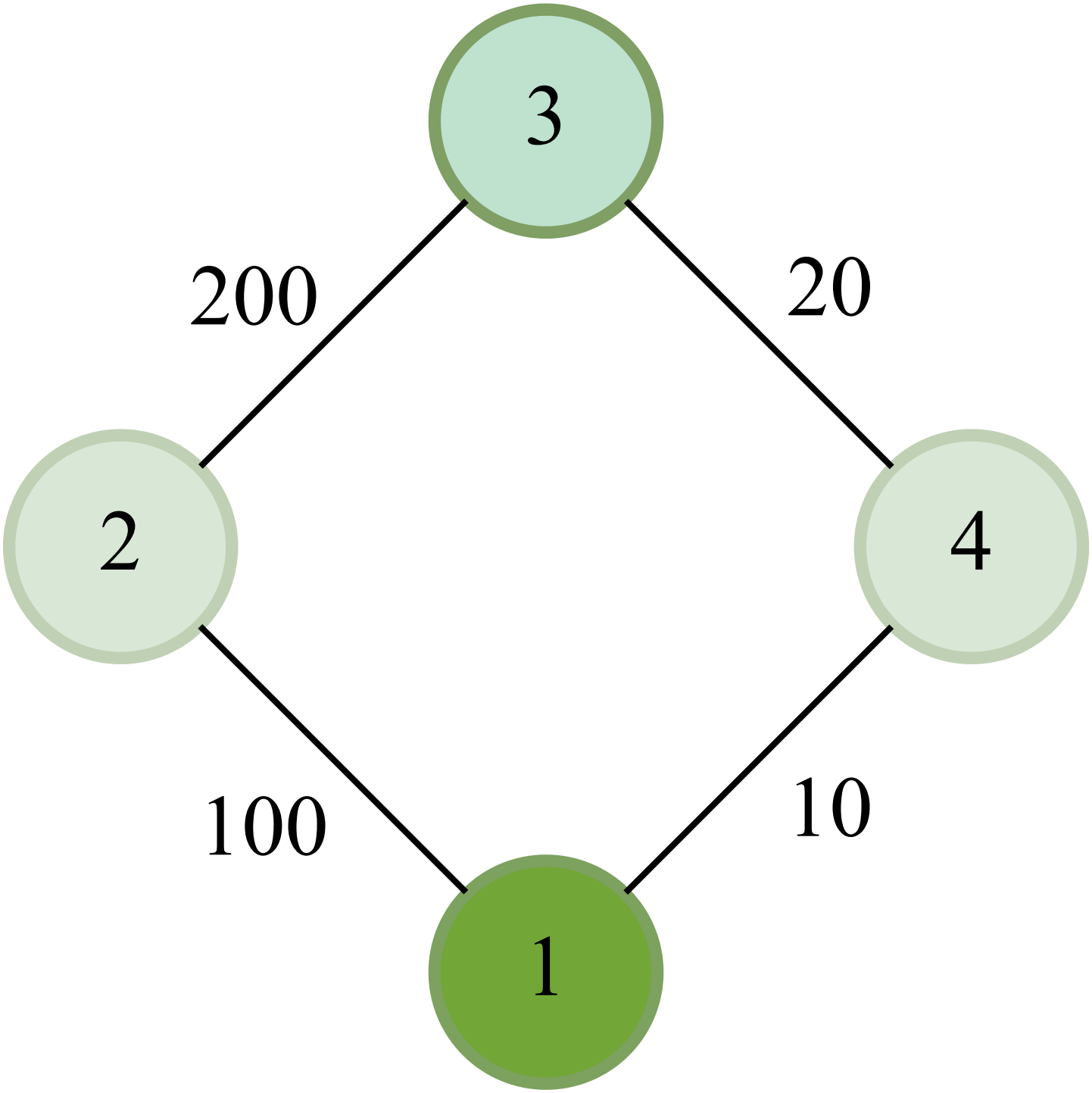
Question description

You have a weighted undirected graph with *graph_nodes* nodes and *m* edges. The stress level of a path between two nodes is defined as the weight of the heaviest edge in that path.

Given a source node *source* and a destination node *destination*, find the minimum possible stress level of a path. If no such path exists, return -1.

Example

Consider the following example below, where *source=1* and *destination=3.*



There are two paths, from node 1 to node 3.

1. The max weighted edge/stress level in path 1 -> 2 -> 3 is 200.
2. The max weighted edge/stress level in the path 1 -> 4 -> 3 is 20.

Return 20, the lower stress level from the second path.

Function Description

Complete the function *getMinimumStress* in the editor with the following parameter(s):

int graph_nodes: number of nodes
int graph_from[m]: one end node for an edge
int graph_to[m]: one end node for an edge
int graph_weight[m]: the weights of the edges
int source: the source node
int destination: the destination node

Returns

int: the value of the least stressful path, or -1 if no path exists

Note: Even though edge endpoints are called *graph_from[i]* and *graph_to[i]*, this is an undirected graph.

Constraints

- $1 \le graph_nodes \le 10^5$
- $1 \le |graph_from| = |graph_to| = |graph_weight| \le 10^5$ *(Where |A| denotes the size of array A)*
- $1 \le graph_from[i], graph_to[i], source, destination \le graph_nodes$
- $0 \le graph_weight[i] \le 10^9$

▼ INPUT FORMAT FOR CUSTOM TESTING

The first line contains two space-separated integers *graph_nodes* and *m,* the number of nodes, and the number of edges of the graph.
Each of the following *m* lines contains three space-separated integers *u, v,* and *w,* meaning there is an edge connecting the nodes *u* and *v* of weight *w.*
The next line contains a single integer, denoting the index of the "*source*" node.
The next line contains a single integer, denoting the index of the "*destination*" node.

▼ SAMPLE CASE 0

Sample Input For Custom Testing

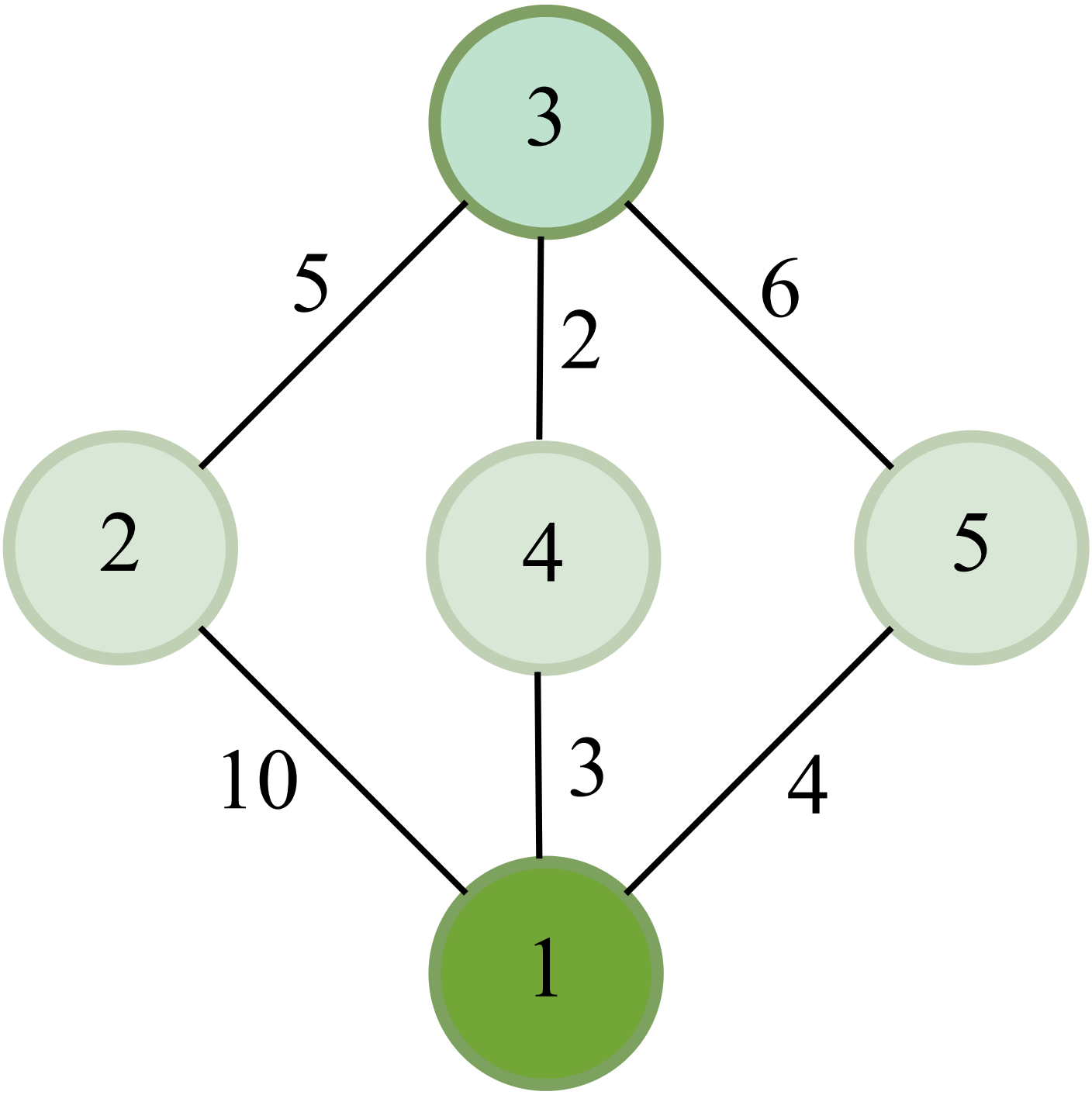
```
STDIN       FUNCTION
-----       --------
5 6         graph_nodes = 5, m = 6
1 2 10      u = 1, v = 2, w = 10
```

```
2 3 5
1 4 3
4 3 2
1 5 4
5 3 6
1          source = 1
3          destination = 3
```

Sample Output

```
3
```

Explanation



There are three paths, from node 1 to node 3 :

1. 1 -> 2 -> 3. Edges have weights 10 and 5, the maximum weighted edge, and the stress level is 10.
2. 1 -> 4 -> 3. Edges have weights 3 and 2, and the stress level = 3.
3. 1 -> 5 -> 3. Edges have weights 4 and 6, and the stress level = 6.

▼ SAMPLE CASE 1

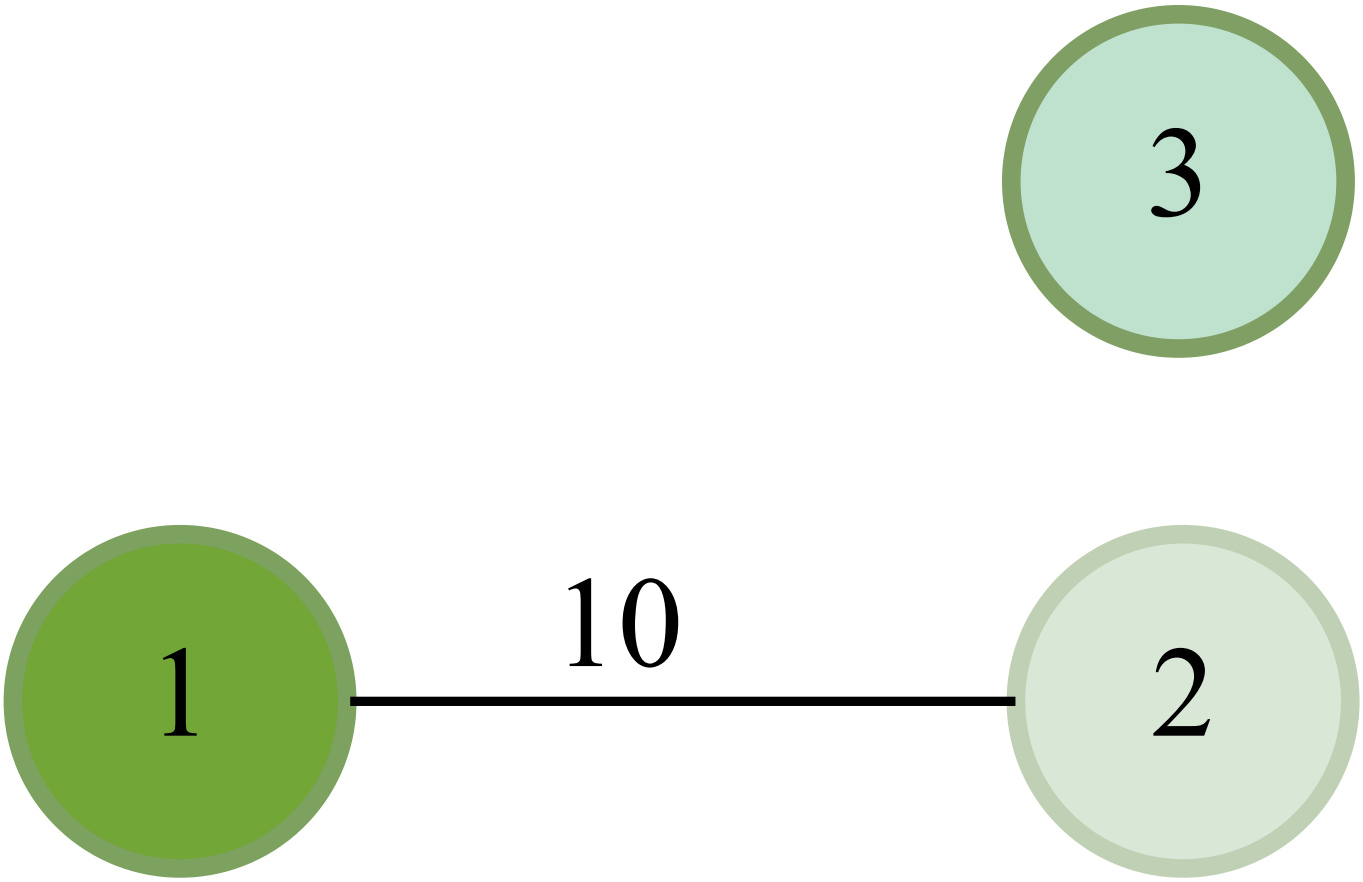
Sample Input For Custom Testing

```
STDIN       FUNCTION
-----       --------
3 1          graph_nodes = 3, m = 1
1 2 10       a = 1, b = 2, w = 10
1           source = 1
3           destination = 3
```

Sample Output

```
-1
```

Explanation



There is no path, from node 1 to node 3.

Interviewer guidelines

▼ SOLUTION

Python Solution

```
sys.setrecursionlimit(10**5)


def getMinimumStress(graph_nodes, graph_from, graph_to, graph_weight, source, destination):

    adj = [[] for i in range(graph_nodes + 1)]
    vis = [False] * (graph_nodes + 1)

    for x, y, w in zip(graph_from, graph_to, graph_weight):
        adj[x].append((y, w))
        adj[y].append((x, w))

    def dfs(node, mx):
        vis[node] = True
        if node == destination:
            return True
        ans = False
        for i, w in adj[node]:
            if w <= mx and not vis[i]:
                ans |= dfs(i, mx)
        return ans

    if not dfs(source, max(graph_weight)):
        return -1

    lo, hi = 0, max(graph_weight)
    while hi - lo > 1:
        vis = [False] * (graph_nodes + 1)
        mid = (hi + lo) // 2
        if dfs(source, mid):
            hi = mid
        else:
            lo = mid
    return hi
```

Candidate's Solution

Language used: **Java 21**

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
```

```java
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;


class Result {

    /*
     * Complete the 'getMinimumStress' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts following parameters:
     *  1. WEIGHTED_INTEGER_GRAPH graph
     *  2. INTEGER source
     *  3. INTEGER destination
     */

    /*
     * For the weighted graph, <name>:
     *
     * 1. The number of nodes is <name>Nodes.
     * 2. The number of edges is <name>Edges.
     * 3. An edge exists between <name>From[i] and <name>To[i]. The weight of the edge is <name>Weight[i].
     *
     */

    public static int getMinimumStress(int graphNodes, List<Integer> graphFrom, List<Integer> graphTo, List<Integer> graphWeight, int source, int destination) {
        Map<Integer, List<int[]>> adj =new HashMap<>();
        for(int i=0;i<graphFrom.size();i++){
            int u =graphFrom.get(i);
            int v=graphTo.get(i);
            int w=graphWeight.get(i);
            adj.computeIfAbsent(u,k-> new ArrayList<>()).add(new int[]{v,w});
            adj.computeIfAbsent(v,k-> new ArrayList<>()).add(new int[]{u,w});
        }

        PriorityQueue<int[]> pq =new PriorityQueue<>(Comparator.comparing(a->a[0]));
        pq.offer(new int[]{0,source});
```

```
        int[] minStress = new int[graphNodes+1];
        Arrays.fill(minStress, Integer.MAX_VALUE);
        minStress[source]=0;

        while(!pq.isEmpty()){
            int[] curr = pq.poll();
            int stress =curr[0];
            int node = curr[1];
            if(node ==destination) return stress;
            if(!adj.containsKey(node)) continue;
            for(int[] nei:adj.get(node)){
                int next = nei[0];
                int w = nei[1];
                int newStress = Math.max(stress, w);
                if(newStress < minStress[next]){
                    minStress[next] = newStress;
                    pq.offer(new int[]{newStress,next});
                }
            }
        }
        return -1;
    }

}
public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        String[] graphNodesEdges = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

        int graphNodes = Integer.parseInt(graphNodesEdges[0]);
        int graphEdges = Integer.parseInt(graphNodesEdges[1]);

        List<Integer> graphFrom = new ArrayList<>();
        List<Integer> graphTo = new ArrayList<>();
        List<Integer> graphWeight = new ArrayList<>();

        IntStream.range(0, graphEdges).forEach(i -> {
            try {
                String[] graphFromToWeight = bufferedReader.readLine().replaceAll("\\s+$", "").split(" ");

```

```
                graphFrom.add(Integer.parseInt(graphFromToWeight[0]));
                graphTo.add(Integer.parseInt(graphFromToWeight[1]));
                graphWeight.add(Integer.parseInt(graphFromToWeight[2]));
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        });

        int source = Integer.parseInt(bufferedReader.readLine().trim());

        int destination = Integer.parseInt(bufferedReader.readLine().trim());

        int result = Result.getMinimumStress(graphNodes, graphFrom, graphTo, graphWeight, source, destination);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
TestCase 0	Easy	Sample	Success	1	0.0773 sec	37.3 KB
TestCase 1	Easy	Sample	Success	1	0.0518 sec	37.8 KB
TestCase 2	Easy	Sample	Success	1	0.0549 sec	37.8 KB
TestCase 3	Easy	Hidden	Success	6	0.6162 sec	122 KB
TestCase 4	Easy	Hidden	Success	6	0.5071 sec	108 KB

TestCase 5	Easy	Hidden	Success	6	0.6894 sec	133 KB
TestCase 6	Easy	Hidden	Success	6	0.6273 sec	146 KB
TestCase 7	Easy	Hidden	Success	6	0.5348 sec	135 KB
TestCase 8	Easy	Hidden	Success	6	0.6397 sec	120 KB
TestCase 9	Easy	Hidden	Success	6	0.6198 sec	126 KB
TestCase 10	Easy	Hidden	Success	6	0.6649 sec	128 KB
TestCase 11	Easy	Hidden	Success	6	0.7753 sec	135 KB
TestCase 12	Easy	Hidden	Success	6	0.4526 sec	93.4 KB
TestCase 13	Easy	Hidden	Success	6	0.3956 sec	82.5 KB
TestCase 14	Easy	Hidden	Success	6	0.7143 sec	138 KB

No comments.

7. Java: DataVille's Optimal Storage

Correct

Coding Greedy Sorting Medium Java

Question description

In the digital city of DataVille, efficient data management is critical. The city's database is represented as a binary array, *databaseStorage*, containing *n* bits of data. Each bit represents one of two types of data:

- Standard data: Represented by 0.
- Priority data: Represented by 1.

To optimize efficiency during peak operational hours, the number of transitions between 0s and 1s in the array must be minimized. Transitions are points in the array where adjacent elements differ (e.g., a 0 followed by a 1 or a 1 followed by a 0).

Operations:

- You can swap any two bits in the array as many times as you like.

The goal is to rearrange the bits to minimize transitions with the minimum number of swaps.

Example

Consider that *databaseStorage* = [1, 0, 1, 0, 1]. One optimal strategy is to swap *databaseStorage[0]* and *databaseStorage[3]*. After performing this operation, the array is [**0**, 0, 1, **1**, 1], which requires only one transition between *databaseStorage[1]* and *databaseStorage[2]*. This requires only 1 swap, so the return value is 1.

Function Description

Complete the function g*etOptimalStorage* in the editor below.

Function Parameters

int databaseStorage[n]: the current organization of the database's storage, with 0s for standard data and 1s for priority data

Returns

int: the minimum number of operations required to minimize transitions between 0s and 1s

Constraints

- $2 \le n \le 10^5$
- *databaseStorage[i]* $\in \{0, 1\}$

INPUT FORMAT FOR CUSTOM TESTING

The first line contains an integer, *n,* the size of *databaseStorage*.

Each of the next lines contains an integer, *databaseStorage[i].*

SAMPLE CASE 0

Sample Input 0

```
STDIN       FUNCTION
-----       --------
7           databaseStorage[] size n = 7
1           databaseStorage = [1, 0, 1, 1, 0, 0, 1]
0
1
1
0
0
1
```

Sample Output 0

```
1
```

Explanation

One optimal strategy is to swap *databaseStorage[1]* (which is 0) with *databaseStorage[6]* (which is 1). After performing this operation, the array becomes [1, **1**, 1, 1, 0, 0, **0**]. This solution requires only 1 swap, so the output is 1.

SAMPLE CASE 1

Sample Input 1

```
STDIN     FUNCTION
-----     --------
9         databaseStorage[] size n = 9
1         databaseStorage = [1, 0, 0, 1, 1, 0, 0, 0, 1]
0
0
1
1
0
0
```

```
0
1
```

Sample Output 1

```
2
```

Explanation

One optimal strategy of operations is:

1. Swap *databaseStorage*[1] and *databaseStorage*[8]. After performing this operation, the array becomes [1, **1**, 0, 1, 1, 0, 0, 0, **0**].
2. Swap *databaseStorage*[2] and *databaseStorage*[4]. After performing this operation, the array becomes [1, 1, **1**, 1, **0**, 0, 0, 0, 0].

Now there is only one transition. The minimum number of swaps is 2.

Interviewer guidelines

▼ SOLUTION

The solution is to count the number of 0s and 1s in the array. Then, we count the number of 0s and 1s that are out of place. The answer is the minimum of the two counts.

```
class Result {

    /*
     * Complete the 'getOptimalStorage' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts INTEGER_ARRAY databaseStorage as parameter.
     */

    public static int getOptimalStorage(List<Integer> databaseStorage) {
        int zeros = 0, ones = 0;

        for (int brick : databaseStorage) {
            if (brick == 0) {
                zeros++;
            } else {
                ones++;
            }
        }
```

```
        int startZero = 0, startOne = 0;

        for (int i = 0; i < ones; ++i) {
            if (databaseStorage.get(i) != 1) {
                startOne++;
            }
        }

        for (int i = 0; i < zeros; ++i) {
            if (databaseStorage.get(i) != 0) {
                startZero++;
            }
        }

        return Math.min(startZero, startOne);
    }
}
```

▼ COMPLEXITY

Time Complexity - O(n)

To traverse over the array, we use a loop that runs n times. The complexity is O(n).

Space Complexity - O(1)

We use only 2 variables, hence the space complexity is O(1).

Candidate's Solution

Language used: **Java 21**

```
import java.io.*;
import java.math.*;
import java.security.*;
import java.text.*;
import java.util.*;
import java.util.concurrent.*;
import java.util.function.*;
import java.util.regex.*;
import java.util.stream.*;
import static java.util.stream.Collectors.joining;
import static java.util.stream.Collectors.toList;

```

```java

class Result {

    /*
     * Complete the 'getOptimalStorage' function below.
     *
     * The function is expected to return an INTEGER.
     * The function accepts INTEGER_ARRAY databaseStorage as parameter.
     */

    public static int getOptimalStorage(List<Integer> databaseStorage) {
        int totalZeros =0;
        int totalOnes =0;
        for(int bit: databaseStorage){
            if(bit == 0){
                totalZeros++;
            }
            else totalOnes++;
        }

        int misplacedOnes =0;
        for(int i=0;i<totalZeros;i++){
            if(databaseStorage.get(i)==1)
            misplacedOnes++;
        }
        int misplacedZeros =0;
        for(int i=0;i<totalOnes;i++){
            if(databaseStorage.get(i)==0)
            misplacedZeros++;
        }
        return Math.min(misplacedOnes, misplacedZeros);
    }

}
public class Solution {
    public static void main(String[] args) throws IOException {
        BufferedReader bufferedReader = new BufferedReader(new InputStreamReader(System.in));
        BufferedWriter bufferedWriter = new BufferedWriter(new FileWriter(System.getenv("OUTPUT_PATH")));

        int databaseStorageCount = Integer.parseInt(bufferedReader.readLine().trim());

        List<Integer> databaseStorage = IntStream.range(0, databaseStorageCount).mapToObj(i -> {
```

```
            try {
                return bufferedReader.readLine().replaceAll("\\s+$", "");
            } catch (IOException ex) {
                throw new RuntimeException(ex);
            }
        })
            .map(String::trim)
            .map(Integer::parseInt)
            .collect(toList());

        int result = Result.getOptimalStorage(databaseStorage);

        bufferedWriter.write(String.valueOf(result));
        bufferedWriter.newLine();

        bufferedReader.close();
        bufferedWriter.close();
    }
}

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
TestCase 0	Easy	Sample	Success	1	0.0465 sec	37.4 KB
TestCase 1	Easy	Sample	Success	1	0.0494 sec	37.5 KB
TestCase 2	Easy	Sample	Success	1	0.0568 sec	37.4 KB
TestCase 3	Easy	Hidden	Success	3	0.0617 sec	37.5 KB
TestCase 4	Easy	Hidden	Success	3	0.0513 sec	37.4 KB
TestCase 5	Easy	Hidden	Success	3	0.0765 sec	37.2 KB

TestCase 6	Easy	Hidden	Success	4	0.0522 sec	37.3 KB
TestCase 7	Easy	Hidden	Success	4	0.0708 sec	37.4 KB
TestCase 8	Easy	Hidden	Success	5	0.0533 sec	37.5 KB
TestCase 9	Easy	Hidden	Success	5	0.0589 sec	37.3 KB
TestCase 10	Easy	Hidden	Success	5	0.055 sec	37.5 KB
TestCase 11	Medium	Hidden	Success	7	0.0617 sec	39.6 KB
TestCase 12	Medium	Hidden	Success	8	0.0776 sec	39.2 KB
TestCase 13	Hard	Hidden	Success	10	0.0782 sec	39.6 KB
TestCase 14	Hard	Hidden	Success	15	0.2623 sec	69.5 KB

No comments.